

# Set-Intersections

## Background

This exercise is modelled on a privacy-preserving data platform that generates insights from multiple data sources without pooling the underlying data in one location or revealing identifiable information. To maintain privacy, only anonymised data is ever stored.

Information is imported as datasets - lists of information about individuals which have been processed to ensure that personal identifying information (PII) has been removed. A dataset has multiple rows of data, with multiple columns. Each row of data has one or more keys which can be used to uniquely identify an individual within the dataset.

One type of key that can be used to identify individuals is UDPRN. UDPRN stands for Unique Delivery Point Reference Number, which is a Royal Mail identifier for every unique delivery address in the UK.

## The Task

Write a program that will allow the comparison of two datasets. These datasets are provided in the form of a CSV file. These simple example CSV files contain a list of UDPRN keys.

The program should perform the following:

- Allow specification of the files to process
- Calculate and display the following:
  - The count of the keys in each file
  - The count of the distinct keys in each file
  - The count of the overlap of all keys between the files (total overlap)
  - The count of the overlap of distinct keys between the two files (distinct overlap)

Total Overlap is defined as the maximum possible overlap of keys between files, e.g.

```
Dataset 1: A B C D D E F F
Dataset 2: A C C D F F F X Y
Distinct Overlap = A C D F = 4
Total Overlap = A C C D D F F F F F F = 11
```

## Additional Considerations

- The program should be able to handle different files in csv format
- Consider how the program would work with larger files both in terms of number of rows of data and number of unique key values
- If approximations are used, ensure the accuracy of the values is appropriately represented
- Feel free to note any assumptions made about this task

## Deliverables

- Source code for the solution
- Instructions on how to build and run
- Appropriate documentation